

Solution to Problem set 4: International Trade I

A.Comparative advantage

1. Explain why companies in developed countries often choose to outsource their call centers to developing countries (in your answer, please refer to Ricardo's Comparative Advantages Theorem).

Ricardo's Comparative Advantages Theorem says that when there are two countries that can trade, and each specializes in the production of a different good (because of lower opportunity costs) then each country should produce the good it is specialized in (and then trade with the other country) – even if one country has an absolute advantage. For instance, if Indian workers are specialized in call centers (either because of experience or because of lower salaries) then trade with the US would lead to outsourcing even if the US would have an absolute advantage.

2. Country A can produce either 2 oranges or 6 lemons. Country B can produce either 2 oranges or 10 lemons. The two countries can trade.
 - a. Which country should produce which product?

Capacity	Oranges	Lemons
Country A	2	6
Country B	2	10

Opportunity costs	To produce 1 oranges	To produce 1 lemon
Country A	3 Lemons	1/3 Orange
Country B	5 Lemons	1/5 Orange

Country A can produce oranges more cheaply (giving up only 3 lemons, compared to 5 lemons for country B) whereas country B can produce lemons more cheaply (giving up 1/5 oranges, compared to 1/3 oranges for Country A).

Hence, although country B has an absolute advantage, it should produce lemons. Country A should produce oranges.

- b. What is the range of prices in which the countries will trade?

The countries will trade if the price of an orange p_o and the price of a lemon p_L satisfy

$$3 \text{ Lemons} > P_o > 5 \text{ Lemons}$$

$$\frac{1}{3} \text{ Orange} > P_L > \frac{1}{5} \text{ Orange}$$

c. Which of the following options is either inefficient or infeasible?

- i. Country A produces only oranges and imports 120 lemons in exchange for 30 oranges.

It is efficient that country A produces oranges. The price here is 4 lemons = 1 orange and $\frac{1}{4}$ orange = 1 lemon. These are within the range above, so this is efficient and feasible.

- ii. Country A produces only oranges and imports 140 lemons in exchange for 40 oranges.

It is efficient that country A produces oranges. The price here is 3.5 lemons = 1 orange and 0.28 orange = 1 lemon. These are within the range above, so this is efficient and feasible.

- iii. Country B produces only lemons and exports 100 lemons in exchange for 25 oranges.

It is efficient that country B produces lemons. The price here is 4 lemons = 1 orange and $\frac{1}{4}$ orange = 1 lemon. These are within the range above, so this is efficient and feasible.

- iv. **Country B produces only lemons and imports 350 oranges in exchange for 100 lemons.**

It is efficient that country B produces lemons. The price here is 1 lemons = 3.5 oranges, which is outside the range. Hence, this is not feasible.

3. Suppose that Germany and Greece that can produce cars and/or airplanes, with the attributes specified in the table below. Assume that there is perfect competition in both markets.
- Show that with these number, Germany *will not export* cars to Greece.
 - The german government wants to *decrease* the hourly wage for workers who are producing cars (without chaning the wage for producing airplanes). How much would the new wage for workers producing cars have to be in order for Germany to start exporting cars to Greece (assume here that productivity stays the same after the change)?

	Germany	Greece
Cars		
Hourly wage	10	5
Hours worked	1	5
Unit costs	10	25
Airplanes		
Hourly wage	10	5
Hours worked	2	15
Unit costs	20	50

(a) As there is perfect competition, prices in each country are equal to the unit costs ($p=c$).

Denote:

- cars as “x” and airplanes as “y”.
- Germany as “country A” and Greece as “country B”.
- Wages as w_i and hours worked on product P as $H_i^P, j \in \{x, y\}$
- Unit costs as $c_i^j = w_i H_i^j$
- The price in which each country is willing to trade as P_i

The formula for the price in country i is then: $P_i = \frac{p_i^x}{p_i^y} = \frac{c_i^x}{c_i^y}, i \in \{A, B\}$

$$P_A = \frac{10}{20} = \frac{1}{2}$$

$$P_B = \frac{25}{50} = \frac{1}{2}$$

[Note: it is sometimes helpful to simplify the table by omitting everything except for unit costs]

Unit costs:

	Germany	Greece
Cars	10	25
Airplanes	20	50

And, one can also translate into the same thought process of comparative advantages we discussed when the table was representing capacity (e.g. in the Robinson & Friday example):

In Germany, the ratio of costs is 10:20 i.e. 1:2. This means that to produce 1 airplane, Germany has to give up the production of 2 cars. In Greece, however, the ratio is 25:50 i.e. 1:2. This means that to produce 1 airplane, Greece has to give up 2 cars, **same as Germany. Thus, no country has a comparative advantage.**

We can see that directly by writing up yet another table:

Unit costs:	To produce 1 plane:	To produce 1 car:
Germany	2 cars	½ plane
Greece	2 cars	½ plane

We can also directly use the formula from the slides : Germany (country A) will export cars to Greece (country B) if and only if $P_A < P_B$. Here, we have $P_A = \frac{1}{2} = P_B = \frac{1}{2}$, so this does not hold.

(b) Assuming that everything stays the same other than the wage, Germany will start exporting cars if the new wage fulfills the following condition:

$$P_A = \frac{w_A * 1}{10 * 2} < P_B = \frac{1}{2}$$

$$2w_A < 20 \Rightarrow w_A < 10$$

This is intuitive: as the countries are the same, Germany can just marginally lower the wage and it already gets a comparative advantage this way.

B.Surplus with trade

1. Suppose that a competitive market opens up for trade and the world price is lower than the existing market price, so that there is import. What happens then to the consumer surplus? What happens to the (local) supplier's surplus?

With import, the consumer surplus increases and the local supplier's surplus decreases.

2. When a competitive market opens up for trade and the world price is higher than the existing market price, so that there is export. What happens to the consumer surplus? What happens to the (local) supplier's surplus?

With export, the local consumer surplus decreases and the local supplier's surplus increases.

3. Explain why a Taarif protects local suppliers.

A taarif reduces the incentive to import (because of the need to pay taxes for imported products), which means the local supplier's surplus will increase, as there will be a larger quantity produced by local suppliers. .